Statistical Research: Research Proposal

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2026-10-06

Motivation

A player's listed height and weight tell us what frame they bring to the rink. They reveal less about how that player contests space: a forward can recover dump-ins, while a defenseman can deny entries, even when neither stands out in the hit column. Jason's (@DBCJason) basketball post on *statistical height* offers a way to think about the difference. By predicting listed height from performance, it describes how tall a player appears to play. We adapt that idea to ask how physical an NHL player's game is for their size.

To address this question, we propose calibrated size above expected (CSAx), which compares the size implied by a player's recorded behavior with the player's listed frame. A higher CSAx describes a more physical recorded profile relative to size; it cannot, by itself, establish courage, overall ability, or success on every play. Its interpretation therefore depends on whether the chosen behaviors reflect physical engagement and whether the profiles agree with evidence beyond the score.

Research questions and design

Because forwards and defensemen encounter different opportunities, we build a separate model for each position. Both draw on hits delivered and received, opponent shots blocked, fights, and recorded contact penalties taken and drawn. For forwards, we add shooting-location and shot-type measures, dump-in recoveries, and forecheck pressure; for defensemen, we add defensive-zone retrievals, botched retrievals, and the share of targeted entries denied. The Data sources report gives the variables and their denominators.

Within each season and position, we combine listed height and weight into a standardized size measure, then use ridge regression to predict it from the recorded behaviors. Five outer assessment folds provide held-out predictions, while five further splits within each training fold select the ridge penalty. Each player-season is scored with a model and frame calibration fitted without that observation. CSAx standardizes the gap between behavior-implied size and the expectation for the player's listed frame within the position and season. Before interpreting rankings, we will compare held-out size prediction with a mean-size baseline and examine residual size trends, annual stability, and the influence of sparse events.

Official NHL draft-year scouting reports published before each player's included NHL seasons offer an earlier view of physical engagement. One rater, blinded to player identity and CSAx, codes whether each report mentions active traits such as checking or puck battles. We will compare player-average CSAx between players whose reports mention these traits and those whose reports do not. A report's silence cannot establish soft play.

We also examine how CSAx relates to the roles teams assign, using official ice-time records to express power-play and penalty-kill minutes as shares of total ice time. The player, teammates, and coaching decisions all help determine these shares. Their associations with CSAx will complement the scouting comparison, although they cannot define toughness on their own.

We next ask how contact changes from the regular season to the postseason. We will compare five-on-five regular-season games excluded from CSAx construction with each team's first four postseason games, giving qualifying teams the same window regardless of how far they advance. Player-season Poisson models will compare delivered-hit counts using ice-time offsets and interactions with CSAx; hits received and opponent shots blocked will add context. We will also examine the full postseason to assess sensitivity to the observation window. Because playoff participation is selective, the analysis concerns players observed in both periods.

Finally, we will ask whether CSAx relates to a player's place in the NHL the following season. We define continuation as at least 300 NHL minutes and fit separate positional logistic models adjusting for listed size, age, current-season games and ice time per game, five-on-five scoring, relative shot attempts, and season. Among players active in the preceding season, we will compare continuation models fitted to the same observations: one uses current-season games and ice time per game, and the other uses the preceding season's values. We will then examine next-season ice time per game among returners. Continuation speaks to roster relevance, while ice time describes the roles of players who return; neither outcome alone verifies what CSAx measures.

Data and feasibility

We join NHL play-by-play, rosters, shifts, and official ice-time records with Corey Sznajder's All Three Zones transition workbook. Our behavioral data cover four regular seasons, 2021–22 through 2024–25, and the roster window extends through 2025–26 for next-season follow-up. The prepared population includes 1,485 forward-seasons and 829 defenseman-seasons, each with at least 300 full-season NHL minutes and 150 matched, tracked five-on-five minutes. Tracking covers games unevenly, and the microstats summarize player-games rather than individual puck battles. Both features shape what we can infer from the scores.

Work plan

Dates (2026)	Planned work
October 6–12	Confirm source matches, exclusions, denominators, and descriptive coverage.
October 13–19	Review positional features, held-out prediction, calibration, and score stability.
October 20–26	Compare scores with scouting descriptions and special-teams roles.
October 27–November 2	Examine postseason engagement and next-season continuation; draft the paper.
November 3–9	Revise the paper and presentation after feedback.
November 10	**Present the completed research.**

Sources

NHL records come through the pinned `nhlscraper` revision. All Three Zones provides the transition workbook and microstat glossary; the source catalog identifies the official NHL scouting reports used for external comparison.